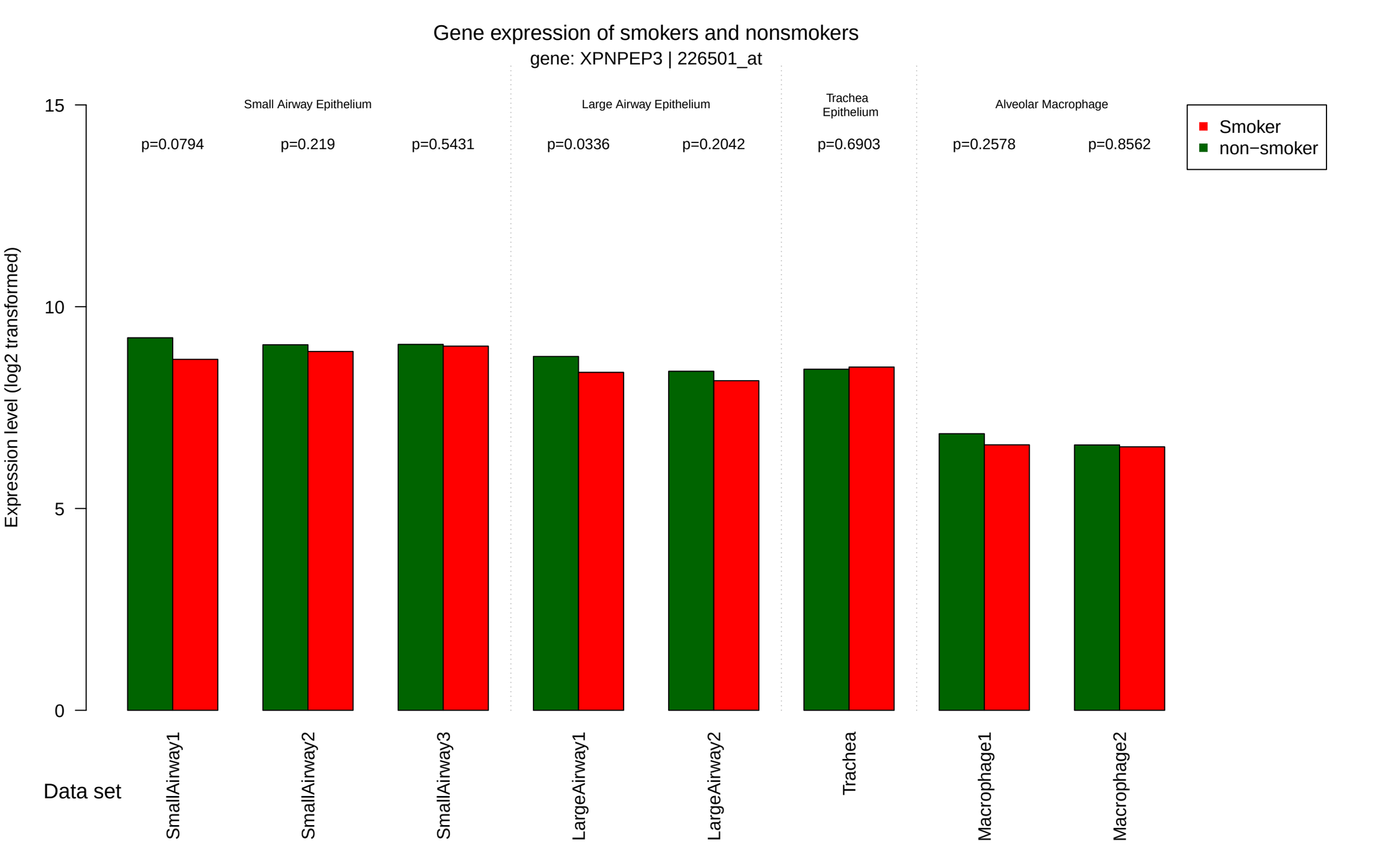
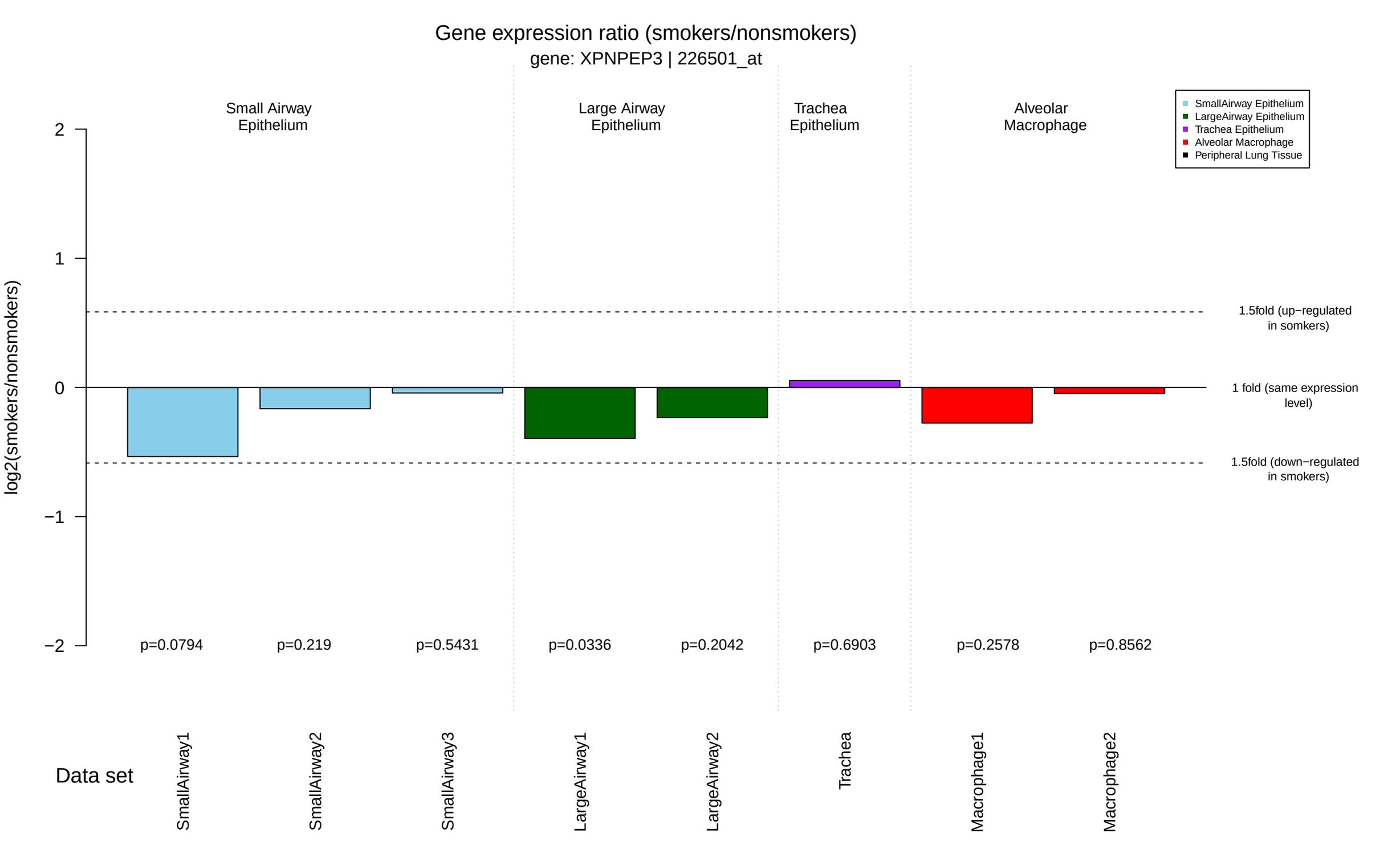
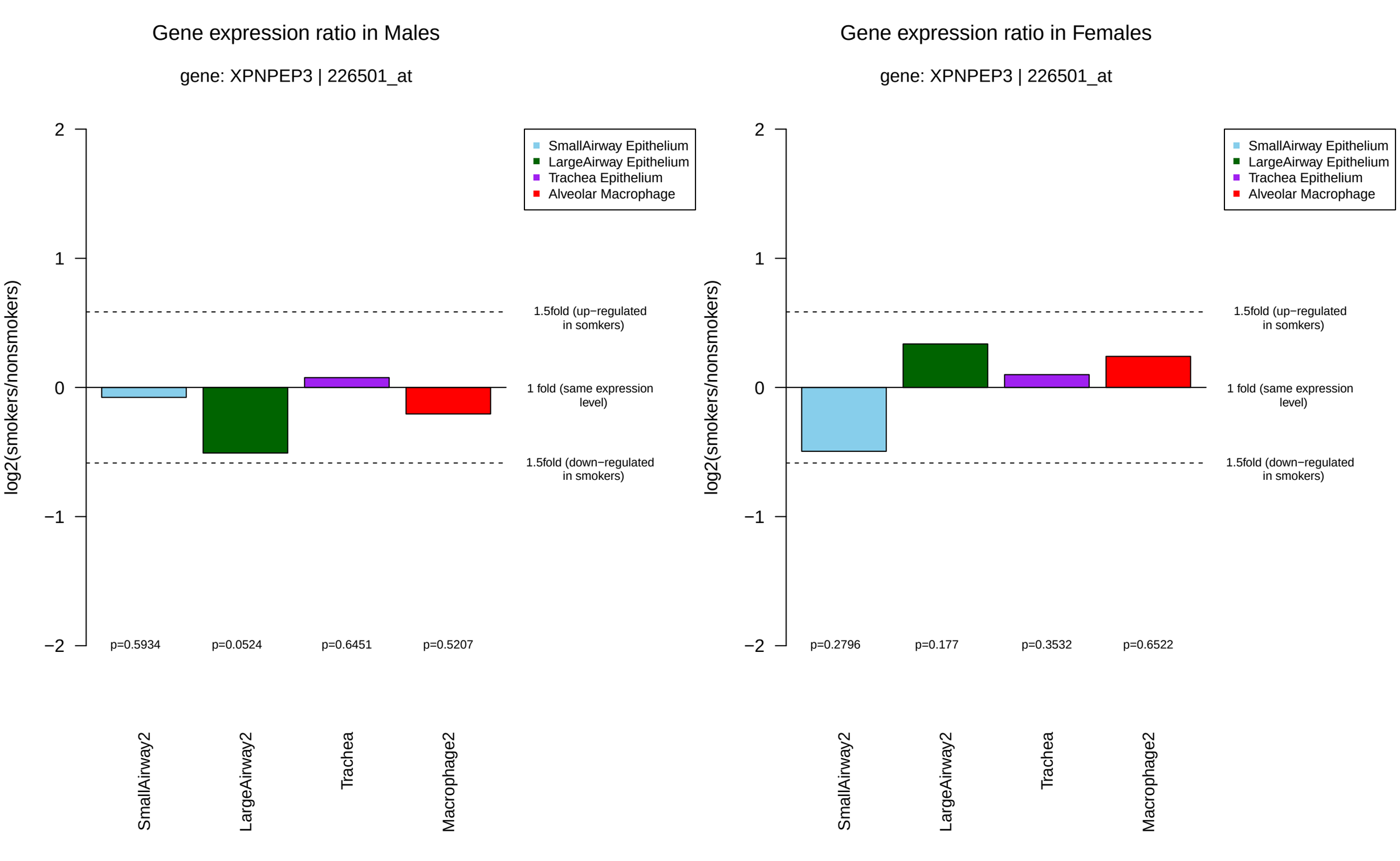
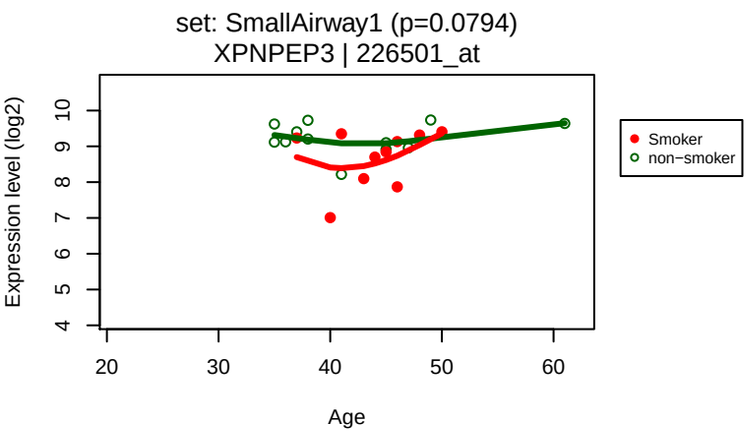
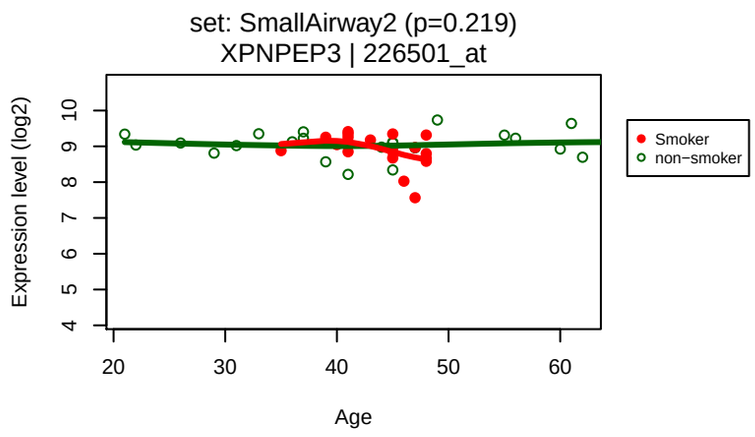
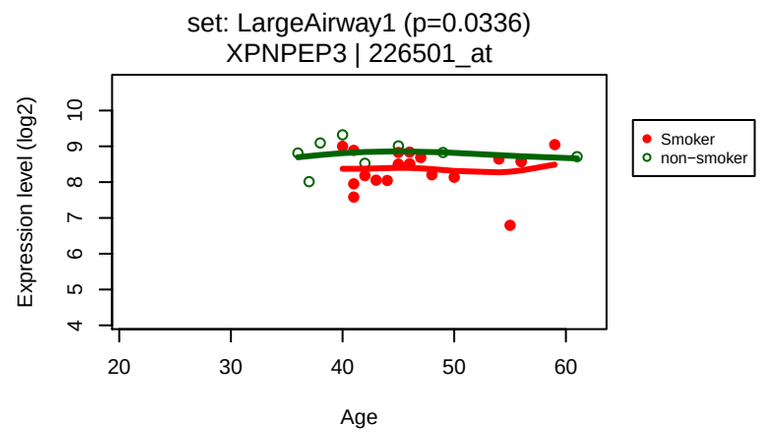

Gene expression of smokers and nonsmokers
gene: XPNPEP3 | 226501_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0794
p=0.219
p=0.5431
p=0.0336
p=0.2042
p=0.6903
p=0.2578
p=0.8562
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

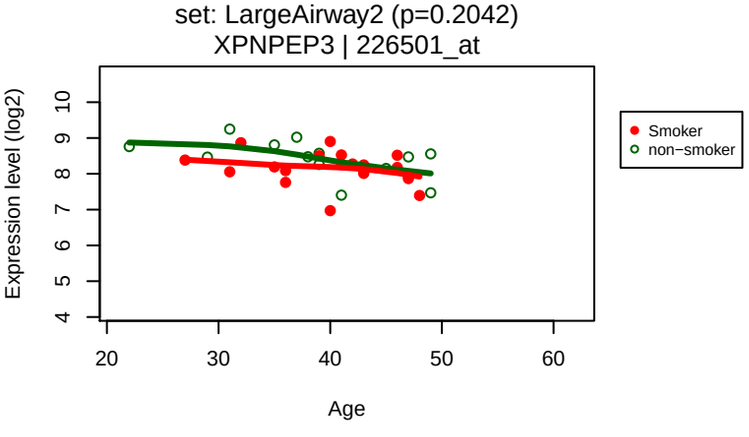

Gene expression ratio (smokers/nonsmokers)
gene: XPNPEP3 | 226501_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0794
p=0.219
p=0.5431
p=0.0336
p=0.2042
p=0.6903
p=0.2578
p=0.8562
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

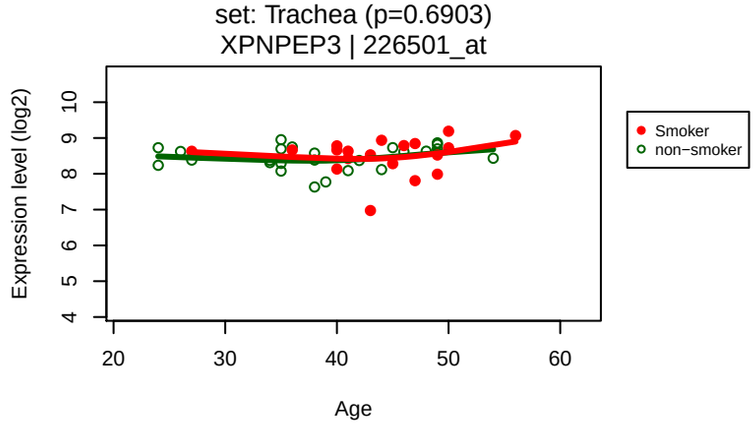

Gene expression ratio in Males
gene: XPNPEP3 | 226501_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5934
p=0.0524
p=0.6451
p=0.5207
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: XPNPEP3 | 226501_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2796
p=0.177
p=0.3532
p=0.6522
SmallAirway2
LargeAirway2
Trachea
Macrophage2

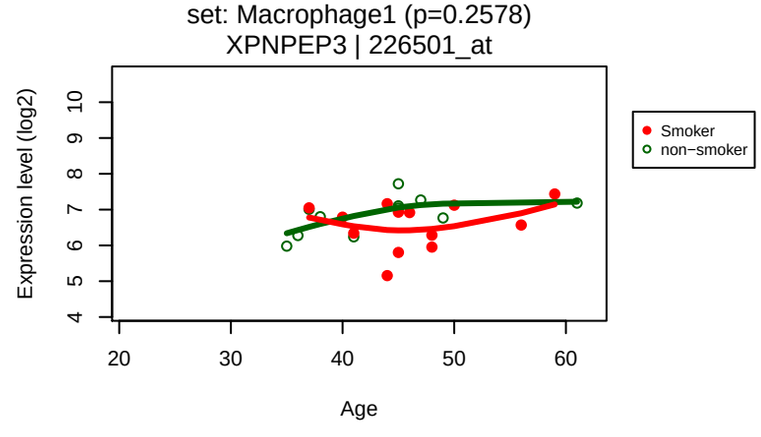

set: SmallAirway1 (p=0.0794)
XPNPEP3 | 226501_at
Expression level (log2)
Age
Smoker
non−smoker

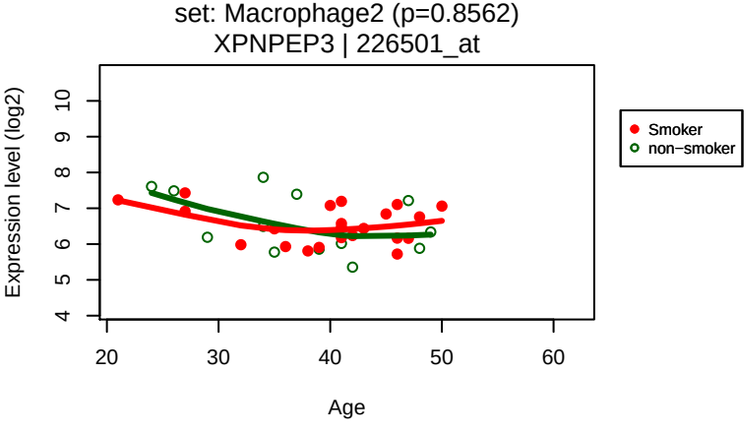

set: SmallAirway2 (p=0.219)
XPNPEP3 | 226501_at
Expression level (log2)
Age
Smoker
non−smoker

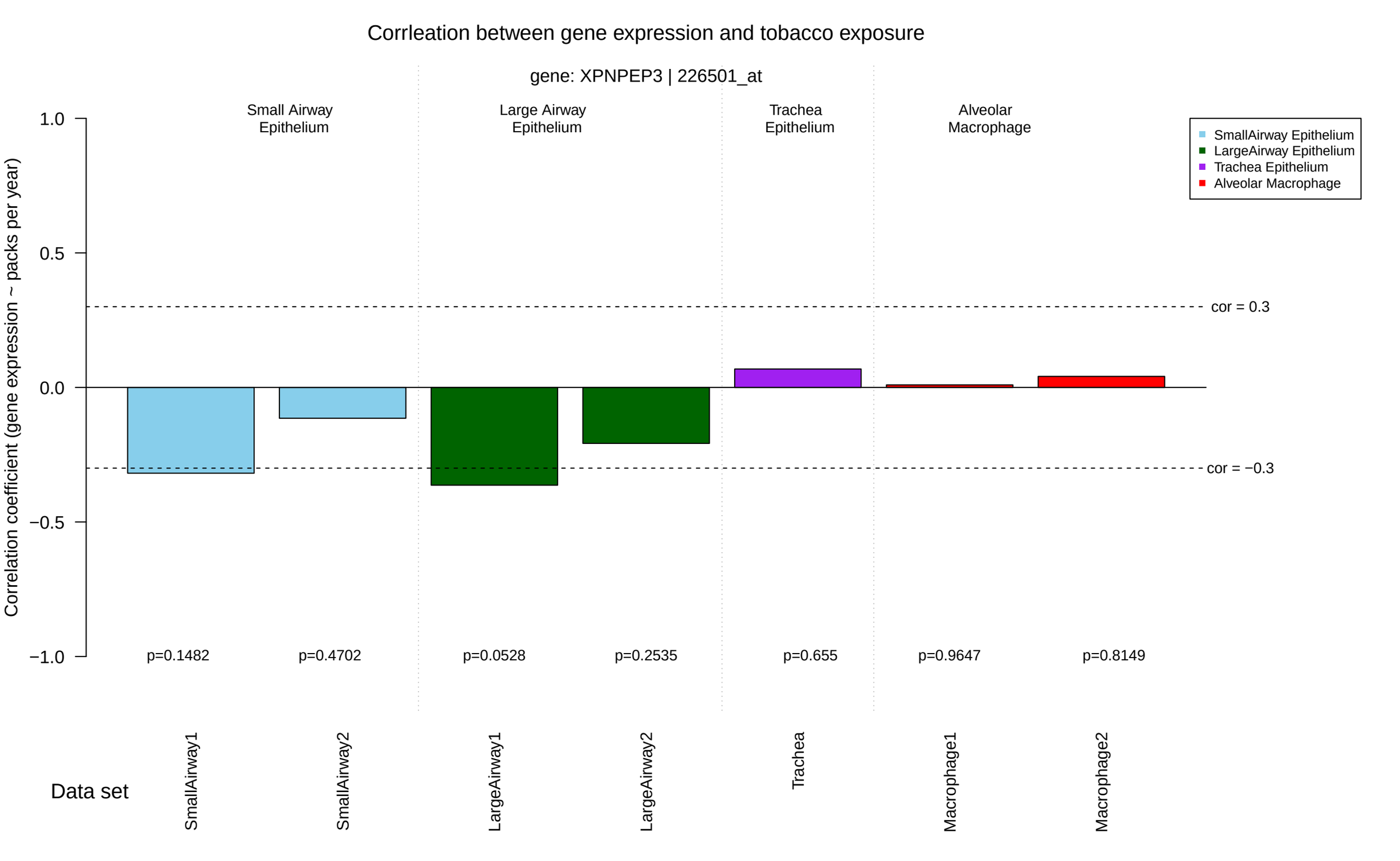

set: LargeAirway1 (p=0.0336)
XPNPEP3 | 226501_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2042)
XPNPEP3 | 226501_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6903)
XPNPEP3 | 226501_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2578)
XPNPEP3 | 226501_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8562)
XPNPEP3 | 226501_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: XPNPEP3 | 226501_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1482
p=0.4702
p=0.0528
p=0.2535
p=0.655
p=0.9647
p=0.8149
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2